

Robust Audio-Visual Generalized Zero-Shot Learning with Quality-Aware Representation Learning

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Abstract

Audio-visual generalized zero-shot learning focuses primarily on transferring knowledge from seen-class features to unseen classes. However, current methods largely ignore modal imbalance and robustness issues in real-world scenarios. Specifically, edge-cloud collaboration has become a mainstream deployment paradigm, where feature transmission is inevitably subject to noise. Additionally, given the intrinsic limitations of modality capacity and the uneven influence of noise across modalities, multimodal training is more likely to give rise to issues such as modality competition. To address these challenges, we propose Quality-Aware Robust Method for Audio-Visual Generalized Zero-Shot Learning. Specifically, we develop an optimal-transport-guided denoising reconstructor. It measures the Wasserstein distance between the latent distributions of clean and corrupted features, and uses the resulting transport plan to align the corrupted distribution with the clean one in the latent space. In addition, under audio-visual feature corruption, existing modality balancing methods for branch learning become substantially less reliable. To address this issue, we propose a semantic-guided diagnostic relearning strategy, which introduces a new purity computation scheme and is better suited to the current setting by incorporating the transferability characteristics of GZSL features. Specifically, we employ HDBSCAN to cluster sub-features by exploring their intrinsic relationships. Extensive experiments show that the proposed method remains highly robust even under extreme noise conditions.

CCS Concepts

• Computing methodologies → Learning latent representations; Computer vision tasks.

Keywords

Generalized Zero-Shot Learning, Modal Imbalance, Optimal-Transport

1 Introduction

Audio-visual generalized zero-shot learning establishes semantic associations between seen and unseen classes, enabling the discriminative knowledge learned from seen-class features to be transferred to unseen classes [29, 45]. However, with the rapid development of mobile devices, Kang et al. [11] introduced the concept of “collaborative intelligence”, which enhances decision-making, problem-solving, and overall performance through cooperation between edge devices and cloud computing. Wireless communication systems constitute an indispensable component of collaborative intelligence, as they support connectivity and data exchange between devices and the cloud. Such technologies [12, 37] enable real-time communication, data transmission, and synchronization. Nevertheless, limited bandwidth and transmission resources may hinder the efficient delivery of large amounts of data between edge devices

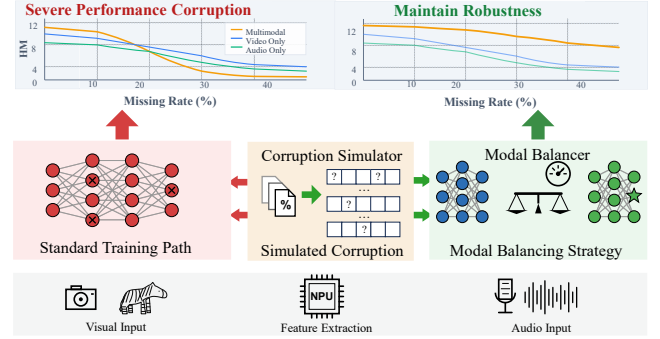


Figure 1: Under feature corruption, models without modality balancing show more severe performance corruption, while the proposed balancing strategy helps preserve robustness.

and the cloud. Moreover, the feature encoding and transmission process must be sufficiently robust to cope with challenges such as channel noise. Specifically, sensing and compression may jointly accumulate distortions in transmitted features, making them approximately subject to Gaussian noise. Under unstable transmission conditions, such as connection fluctuations and packet loss, features may also suffer from data missing. As a result, the features transmitted to the cloud may lose reliability and completeness. This gap in “information quality” can further aggravate modality imbalance.

As illustrated in Fig. 1, models trained with modality balancing exhibit stronger robustness and are less likely to fail when the input data become incomplete. Existing modality balancing methods can be broadly categorized into loss reweighting strategies [30], gradient-modulation-based dynamic balancing strategies [24, 33], and learning-state diagnosis and relearning strategies [34]. Their common objective is to suppress modality dominance during joint training and reduce optimization imbalance across different branches. However, most of these methods are designed for conventional multimodal discriminative learning settings, where balancing signals are derived from relatively stable feedback, gradient contributions, or branch states. When the inputs are incomplete or severely corrupted, the reliability of this assumption is substantially weakened. Under severe corruption, these cues may fluctuate considerably and become unreliable, which in turn makes the balancing process itself unstable. Among these approaches, diagnosis-and-relearning strategy are relatively more suitable for audio-visual generalized zero-shot learning, because they focus more directly on the learning states of individual branches. This property makes them more appropriate as a basis for cross-modal collaborative modeling. Nevertheless, such methods still rely on the current representations to reflect the underlying semantic

structure. Under severe corruption, the boundaries between different classes may become blurred, and the discriminative semantic space can no longer serve as an effective signal to guide subsequent tasks. In this case, representation-based state diagnosis becomes unreliable, and the subsequent relearning process consequently loses a trustworthy basis. Therefore, achieving effective balancing in corrupted audio-visual generalized zero-shot learning requires a more structure-aware diagnostic signal. Such a signal should remain effective under extreme noise conditions while being more sensitive to the learning efficiency of different branches.

To address the above challenges, we propose Quality-Aware Robust Method (QARM) for corrupted audio-visual generalized zero-shot learning. QARM achieves this goal through improved modality balancing and stable denoising reconstruction. Specifically, to mitigate the effects of noise corruption, feature missingness, and transmission distortion in edge-cloud collaborative scenarios, we design an Optimal-Transport-guided Denoising Reconstructor (OTDR) to reduce the interference of corrupted features with subsequent cross-modal modeling. This module adopts a denoising-and-reconstruction paradigm with mask prediction and residual refinement to improve reconstruction stability. To avoid the inefficiency of heavily stacked multilayer perceptrons, we employ implicit neural representations (INRs) to complete missing information more effectively. In addition, optimal transport theory and the Wasserstein distance are introduced to constrain the latent representations, which preserving the geometric integrity of reconstructed features and alleviating representation drift. To further relieve modality imbalance under severe corruption, we develop a Semantic-Guided Diagnostic Relearning (SGDR) strategy to model the learning state of each modality branch. Specifically, while HDBSCAN [2, 18] is originally intended to reduce the influence of noise on clustering. While we observe that, in GZSL, some feature dimensions do not effectively convey information, and samples with similar features do not necessarily share the same set of sub-feature patterns. Motivated by this observation, we propose a sub-feature clustering scheme. More specifically, we first use HDBSCAN to identify potential intra-class sub-feature clusters, and then compute the intersection-over-union (IoU) of sub-feature cluster responses among samples from the same class as a branch-level learning indicator. Based on this indicator, the method can assess branch-wise learning states at a finer granularity, which enabling more precise regulation of modal balance during training and improving the robustness of cross-modal knowledge transfer.

The main contributions of this paper are as follows:

- We propose Optimal-Transport-guided Denoising Reconstructor, which combines INR with optimal transport constraints for effective reconstruction and denoising.
- We design a modality-contribution evaluation method. Specifically, an HDBSCAN-based sub-feature clustering strategy is introduced to reduce the interference of noise during clustering while suppressing the influence of less informative dimensions in the original feature space. This design enables a more reliable fine-grained evaluation.
- Extensive experiments show that our method remains competitive under the standard setting and achieves stronger robustness under corrupted conditions.

2 Related Work

2.1 Audio-visual Generalized Zero-shot Learning

Audio-visual generalized zero-shot learning aims to transfer knowledge from seen classes to unseen ones by exploiting the associations between audio-visual modalities [17, 20, 21, 35, 46]. Existing methods improve this process mainly through enhanced cross-modal interaction, stronger dynamic representation modeling, and refined semantic alignment. Specifically, TCAF [19] improves local audio-visual correspondence through temporal cross-modal attention. Beyond such interaction modeling, MDFT [15] further strengthens dynamic representation learning by employing a motion-disentangled spiking Transformer. Moving from representation modeling to semantic transfer, EZ-AVGZL [22] improves the transferability of shared representations through a simpler text-audio-visual alignment framework. However, these methods are mostly developed under the assumption of relatively ideal input features and pay limited attention to representation corruption caused by noise contamination, feature missingness, and transmission distortion in real-world edge-cloud collaborative scenarios. Therefore, we propose QARM for robust modeling under corrupted inputs.

2.2 Feature Denoising and Reconstruction

In real-world deployment environments, various noise can directly degrade intermediate features and impair subsequent recognition, fusion, and knowledge transfer [43]. Therefore, corrupted feature recovery has gradually progressed from explicit denoising, to continuous completion, and further to distribution-constrained reconstruction. Early methods mainly relied on residual learning or encoder-decoder architectures for denoising and local recovery [44]. More recent studies introduced implicit neural representations to model recovery as continuous function fitting, which improving completion flexibility [4, 26, 28]. Building on this line, optimal transport has been further employed to constrain recovered features at the distribution level, which helps reduce representation drift and preserve global geometric structure [3, 32]. Motivated by this progression, we propose OTDR, which integrates differential denoising, INR-based completion, and optimal transport regularization for more robust corrupted feature reconstruction.

2.3 Modality Imbalance

During joint optimization, deep networks tend to favor modalities that are more informative or easier to learn, often leading to imbalanced contributions across branches [9, 14, 25, 31, 39, 47]. To alleviate this issue, existing studies balance modality contributions by adjusting gradient magnitudes during backpropagation. For example, OGM-GE [24] dynamically modulates unimodal gradients based on training performance, while OPM-GE [33] further suppresses modality dominance by randomly dropping part of the features. PMR [5] alleviates premature convergence by introducing prototype-based entropy regularization, and other methods promote finer-grained cooperation through sample-level modality valuation. Although effective in general multimodal tasks, these methods mainly rely on superficial training signals and are insufficient for characterizing branch-wise learning differences driven by

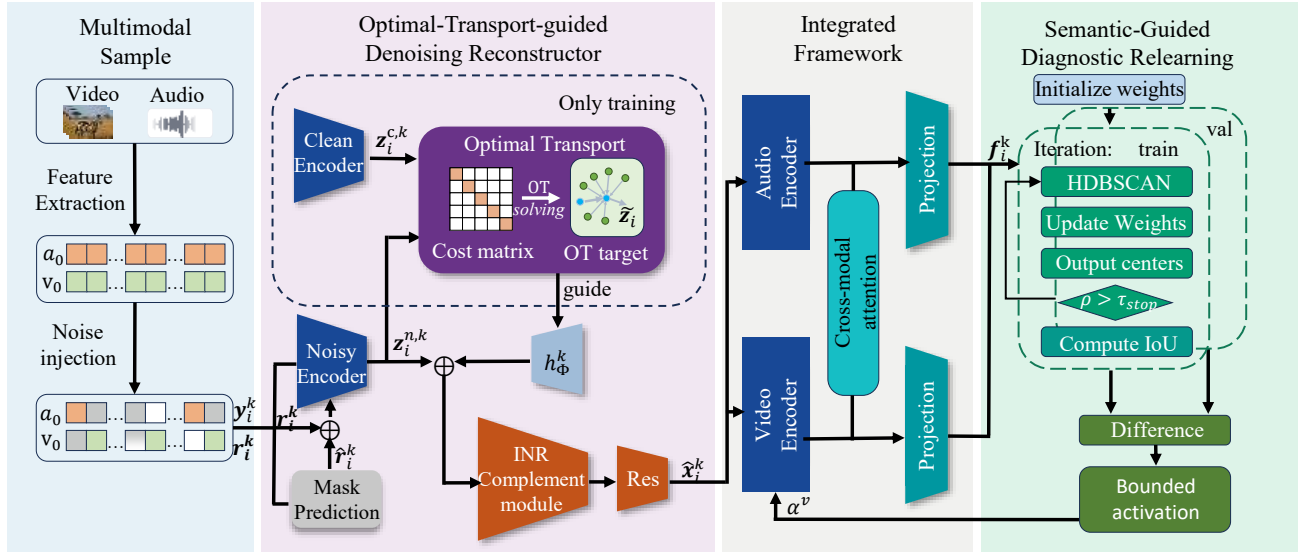


Figure 2: Overall pipeline of QARM. Given extracted audio and visual features, corrupted perturbations are first introduced to simulate noisy conditions. OTDR then restores robust features through mask prediction, dual-branch latent encoding, optimal-transport-guided correction, INR-based completion, and residual refinement. The recovered features are subsequently fed into the multimodal interaction backbone. Meanwhile, SGDR performs iterative sub-feature discovery and IoU-based diagnostic evaluation on branch features to generate control signals for branch-wise regulation during training.

semantic transfer in GZSL. In our work, we propose SGDR, which uses an evaluation metric better aligned with the feature transfer characteristics of GZSL as the control signal to achieve fine-grained regulation of different learning branches.

3 Preliminary

In audio-visual generalized zero-shot learning, the extracted audio and visual features of the i -th sample are denoted by $\mathbf{a}_i$ and $\mathbf{v}_i$, respectively, and the semantic prototype of class c is denoted by $\mathbf{s}_c$. Let $\mathcal{Y}^s$ and $\mathcal{Y}^u$ represent the seen and unseen label sets, respectively, where $\mathcal{Y}^s \cap \mathcal{Y}^u = \emptyset$. The training set is constructed only from seen classes, while the test set contains samples from both seen and unseen classes. Under corrupted deployment conditions, the corrupted input feature of modality $k \in \{a, v\}$ is denoted by $\mathbf{y}_i^k$.

4 Method

The overall pipeline of the proposed Quality-Aware Robust Method (QARM) framework is illustrated in Fig. 2. Given an audio-visual sample, we first extract the audio and visual features and inject corruption to simulate corrupted conditions. The corrupted features are then fed into the Optimal-Transport-guided Denoising Reconstructor (OTDR), where mask prediction, noisy feature encoding, clean latent guidance, optimal-transport-based correction, INR-based completion, and residual refinement are jointly performed to recover more reliable modality representations. The reconstructed audio and visual features are subsequently sent to the modality-specific encoders, followed by cross-modal attention and semantic projection for multimodal representation learning. During this process, the Semantic-Guided Diagnostic Relearning (SGDR) strategy

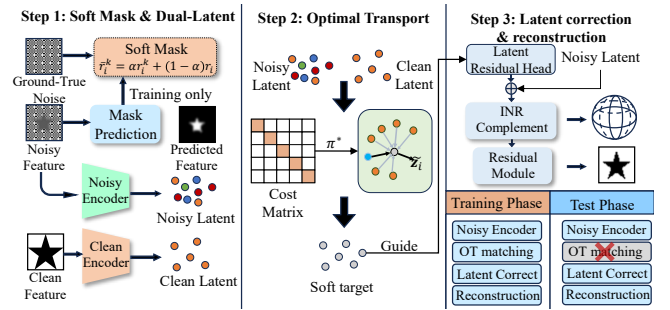


Figure 3: Framework of the proposed Optimal-Transport-guided Denoising Reconstructor.

is further introduced to diagnose branch-wise learning states and generate control signals, which are used to regulate the optimization of the audio and visual encoders during training. In this way, OTDR mainly improves the quality of corrupted input representations before multimodal interaction, while SGDR further stabilizes the subsequent joint optimization process, and the two modules together form a coordinated robust learning framework for corrupted audio-visual generalized zero-shot learning.

4.1 Optimal-Transport-guided Denoising Reconstructor

OTDR aims to recover a robust representation through denoising and completion before downstream multimodal learning. It first encodes corrupted inputs into a noisy latent space with soft

mask guidance, then introduces clean latent references and optimal transport to correct the latent geometry, and finally generates the recovered feature via INR-based completion.

For each corrupted input y_i^k , we first estimate a soft corruption mask and then feed the corrupted feature together with the training-stage soft mask into the noisy encoder. These steps are written as

$$\begin{aligned} \hat{r}_i^k &= M^k(y_i^k), \\ \bar{r}_i^k &= \lambda r_i^k + (1 - \lambda) \hat{r}_i^k, \\ z_i^{n,k} &= D_\theta^{n,k}([y_i^k, \bar{r}_i^k]), \end{aligned} \quad (1)$$

where $M^k(\cdot)$ denotes the mask predictor, $\bar{r}_i^k$ is the training soft mask, and $D_\theta^{n,k}(\cdot)$ denotes the noisy encoder. The coefficient λ gradually decreases over training stages, such that the model transitions from stronger ground-truth mask guidance to greater reliance on the predicted mask. During inference, we directly set $\bar{r}_i^k = \hat{r}_i^k$.

Since the noisy latent may still contain corruption-induced bias, we further refine it with a residual latent head. Meanwhile, the clean feature is mapped to an auxiliary clean latent space to provide a reference distribution during training:

$$z_i^{'k} = z_i^{n,k} + h_\phi^k(z_i^{n,k}), \quad z_i^{c,k} = G_\psi^k(x_i^k), \quad (2)$$

where $h_\phi^k(\cdot)$ denotes the latent residual head, $\gamma \in (0, 1)$ is a learnable scaling factor, and $G_\psi^k(\cdot)$ denotes the auxiliary clean encoder.

To align the refined noisy latent with the clean latent geometry, we impose an entropic optimal transport constraint in the normalized latent space. Specifically, we first normalize both latent representations and define the pairwise transport cost:

$$\bar{z}_i^{'k} = \frac{z_i^{'k}}{\|z_i^{'k}\|_2}, \quad \bar{z}_j^{c,k} = \frac{z_j^{c,k}}{\|z_j^{c,k}\|_2}, \quad C_{ij}^k = \|\bar{z}_i^{'k} - \bar{z}_j^{c,k}\|_2^2. \quad (3)$$

Based on the cost matrix, the optimal transport plan and the corresponding barycentric soft target are jointly defined as

$$\begin{aligned} \pi^{\star,k} &= \arg \min_{\pi \in \Pi(a,b)} \langle \pi, C^k \rangle + \varepsilon \sum_{i,j} \pi_{ij} \log \pi_{ij}, \\ \bar{z}_i^k &= \frac{\sum_j \pi_{ij}^{\star,k} \bar{z}_j^{c,k}}{\sum_j \pi_{ij}^{\star,k}}. \end{aligned} \quad (4)$$

Here, $\Pi(a, b)$ denotes the set of transport plans satisfying the prescribed marginals, and $\bar{z}_i^k$ is the OT-induced barycentric soft target. The resulting OT regularization is formulated as

$$\mathcal{L}_{OT}^k = \sum_{i,j} \pi_{ij}^{\star,k} C_{ij}^k + \frac{1}{B} \sum_i \|\bar{z}_i^{'k} - \text{sg}(\bar{z}_i^k)\|_2^2, \quad (5)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. The first term encourages distribution-level transport alignment, while the second term further pulls each refined noisy latent toward its barycentric clean target at the sample level.

After latent refinement, the recovered feature is generated by the INR completion module conditioned on the refined latent:

$$x_{\text{inr},i}^k = F_\eta^k(z_i^{'k}), \quad \hat{x}_i^k = \text{Res}^k(x_{\text{inr},i}^k), \quad (6)$$

where $x_{\text{inr},i}^k$ denotes the direct INR output and is also used in the intermediate reconstruction loss, while $\hat{x}_i^k$ denotes the final recovered feature used by the downstream multimodal framework.

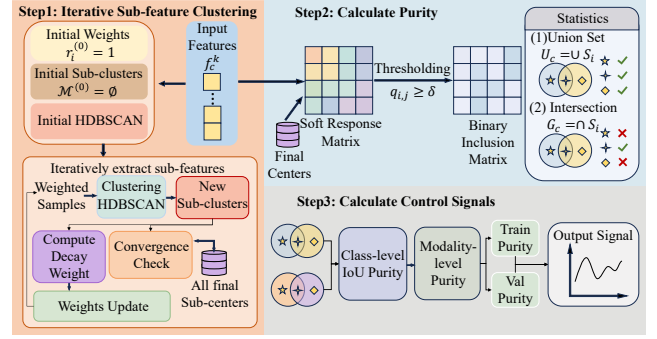


Figure 4: Framework of the proposed Semantic-Guided Diagnostic Relearning, where iterative sub-feature and purity estimation are used to generate diagnostic signals.

To jointly preserve reconstruction fidelity, directional consistency, discriminative alignment, mask prediction quality, and distributional stability, we define the main objective as

$$\begin{aligned} \mathcal{L}_{\text{main}} &= 0.20 \mathcal{L}_{\text{mse}}(\hat{x}_i^k, x_i^k) + 0.20 \mathcal{L}_{\text{mse}}(x_{\text{inr},i}^k, x_i^k) \\ &\quad + 0.15 \mathcal{L}_{\text{cos}}^k + 0.10 \mathcal{L}_{\text{sep}}^k \\ &\quad + 0.15 \mathcal{L}_{\text{bce}}^k + 0.20 \mathcal{L}_{\text{cov}}^k, \end{aligned} \quad (7)$$

where the coefficients sum to 1. The total loss is then given by

$$\mathcal{L}_{\text{total}}^k = \mathcal{L}_{\text{main}}^k + \mathcal{L}_{\text{OT}}^k. \quad (8)$$

By jointly optimizing the above terms, OTDR improves both reconstruction quality and latent-space geometric robustness under corrupted conditions.

4.2 Semantic-Guided Diagnostic Relearning

Although OTDR improves corrupted inputs, multimodal joint optimization may still suffer from branch-wise imbalance. A well-learned branch should yield relatively consistent and class-transferable sub-feature activations for samples from the same class. Based on this intuition, we use purity to measure class-wise shared sub-feature consistency rather than conventional clustering compactness or separation. In this way, purity serves as a branch-wise structural consistency indicator for diagnostic relearning.

For modality k and class c , we first collect the class-wise feature set

$$\mathcal{F}_c^k = \{f_i^k \mid y_i = c\}. \quad (9)$$

SGDR then performs iterative sub-feature discovery within each class. At iteration t , we sample class features according to the residual weights and apply HDBSCAN to discover a set of sub-feature clusters. For each discovered sub-feature cluster, its center is computed in the original feature space and denoted by $\mu_j^{k,c}$. Based on the samples assigned to this sub-feature cluster, we further construct a dimension mask $\omega_j^{k,c}$ by suppressing unstable high-variance dimensions. The newly discovered sub-feature clusters are then added into the class-wise sub-feature cluster set

$$\mathcal{M}_c^k \leftarrow \mathcal{M}_c^k \cup \{(\mu_j^{k,c}, \omega_j^{k,c})\}. \quad (10)$$

Given the discovered sub-feature clusters at iteration t , we evaluate how well each sample is explained by the current sub-feature

cluster set. Specifically, the maximum explained degree of sample i is defined as

$$e_i^{(t)} = \max_j q_{i,j}^{k,c}, \quad (11)$$

where $q_{i,j}^{k,c}$ denotes the soft response of sample i to the j -th sub-feature cluster. Based on the current explained degree, we first update the residual weight of each sample, and then derive a normalized selection probability $p_i^{(t+1)}$ from the updated residual weights:

$$r_i^{(t+1)} = r_i^{(t)} (1 - \alpha_{\text{res}} e_i^{(t)}), \quad p_i^{(t+1)} = \frac{\max(r_i^{(t+1)}, \rho_{\min})}{\sum_m \max(r_m^{(t+1)}, \rho_{\min})}. \quad (12)$$

where $p_i^{(t+1)}$ denotes the normalized selection probability of sample i at iteration $t+1$, and $\rho_{\min}$ is a lower bound introduced to prevent extremely small residual weights from vanishing numerically.

To avoid repeatedly extracting highly similar sub-feature clusters, we measure the overlap between newly discovered centers and previously discovered ones:

$$\rho^{(t)} = \frac{1}{J_t} \sum_{j=1}^{J_t} \max_{j'} \cos(\mu_j^{k,c,(t)}, \mu_{j'}^{k,c,(\text{old})}), \quad (13)$$

where J_t denotes the number of newly discovered sub-feature cluster centers at iteration t . The iterative discovery stops when

$$\rho^{(t)} \geq \rho_{\text{stop}} \quad (14)$$

or when the maximum iteration number is reached. After convergence, the resulting class-wise sub-feature cluster set is written as

$$\mathcal{M}_c^k = \{(\mu_1^{k,c}, \omega_1^{k,c}), \dots, (\mu_{J_c}^{k,c}, \omega_{J_c}^{k,c})\}. \quad (15)$$

For each sample, we compute its masked soft response to the discovered sub-feature clusters as

$$q_{i,j}^{k,c} = \frac{\exp(\tau \text{sim}(\mathbf{f}_i^k, \mu_j^{k,c}; \omega_j^{k,c}))}{\sum_{j'} \exp(\tau \text{sim}(\mathbf{f}_i^k, \mu_{j'}^{k,c}; \omega_{j'}^{k,c}))}, \quad (16)$$

where $\text{sim}(\cdot)$ denotes the masked cosine similarity under the corresponding dimension mask, and τ is a temperature factor. We then obtain the binary activation and the activated sub-feature set by

$$b_{i,j}^{k,c} = \mathbf{1}(q_{i,j}^{k,c} \geq \delta), \quad S_i^{k,c} = \{j \mid b_{i,j}^{k,c} = 1\}, \quad (17)$$

where δ is the binarization threshold.

Based on the activation patterns, we define the total activated sub-feature union and the shared sub-feature set of class c as

$$U_c^k = \left\{j \mid \sum_i b_{i,j}^{k,c} > 0\right\}, \quad G_c^k = \left\{j \mid \frac{1}{N_c} \sum_i b_{i,j}^{k,c} \geq \eta\right\}, \quad (18)$$

where N_c is the number of samples in class c , and η is the shared-sub-feature threshold. The class-wise purity and modality-level purity are then defined as

$$p_c^k = \frac{|G_c^k|}{|U_c^k| + \epsilon}, \quad p^k = \frac{\sum_c N_c p_c^k}{\sum_c N_c}. \quad (19)$$

To transform the train-validation purity discrepancy into a reset control signal, we define

$$\alpha_{\text{raw}}^k = \tanh(\max(0, P_{\text{val}}^k - P_{\text{train}}^k)), \quad a_t^k = w(t) \alpha_{\text{raw}}^k, \quad (20)$$

Algorithm 1 Semantic-Guided Diagnostic Relearning

Require: Training set $\mathcal{D}$, validation set $\mathcal{V}$, total epoch number T , branch parameters Θ_k , $k \in \{a, v\}$, diagnostic frequency H

- 1: **for** $t = 0, \dots, T-1$ **do**
- 2: Update model parameters by standard training
- 3: **if** $t \bmod H = 0$ **then**
- 4: **for** each modality $k \in \{a, v\}$ **do**
- 5: Extract training feature set and validation feature set for branch k
- 6: **for** each class c **do**
- 7: Build class-wise feature set $\mathcal{F}_c^k$ according to Eq. (9)
- 8: Iteratively discover sub-feature clusters and update residual weights according to Eqs. (10)–(14)
- 9: **end for**
- 10: Compute P_{train}^k according to Eqs. (16)–(19)
- 11: Repeat the same procedure on the validation split to obtain P_{val}^k
- 12: Compute D^k , α_{raw}^k , and a_t^k according to Eq. (20)
- 13: Update branch parameters Θ_k according to Eq. (21)
- 14: **end for**
- 15: **end if**
- 16: **end for**

where $w(t)$ is a stage-dependent decay factor. The branch parameters are then softly reset by

$$\Theta_t^k \leftarrow a_t^k \Theta_{\text{init}}^k + (1 - a_t^k) \Theta_{\text{opt}}^k, \quad (21)$$

where a_t^k denotes the branch-wise reset intensity for modality k at epoch t . When $a_t^k = 0$, the current optimized parameters are fully retained; a larger a_t^k pulls the branch parameters closer to their initialization state, yielding a stronger soft reset for subsequent relearning. Θ_{init}^k and Θ_{opt}^k denote the initialized and current optimized parameters of branch k , respectively.

In this way, SGDR provides a GZSL-oriented diagnostic signal for branch-wise relearning, while OTDR improves the robustness of the input representations before multimodal interaction.

5 Experiments

5.1 Implementation Details

In this paper, we train and access our proposed model in (G)ZSL scenarios on single NVIDIA 5090 GPU using the VGGSound [1], UCF101 [27], and ActivityNet [8]. The dimensions of features are set to input = 512, hidden = 512, output = 300. The corresponding text features are extracted with Word2vec pretrained model. For noise intensity, AWGN intensity is set to 0.5/0.6/0.7/0.8/0.9, data missing rate is set to 0.1/0.2/0.3/0.4, bit-flip rate is set to 0/0.1/0.2.

5.2 Corruption Definition

To simulate transmission-induced corruption, we perturb both audio and visual features after feature extraction and before fusion. Let $\mathbf{x} \in \mathbb{R}^d$ denote the transmitted feature vector and $\tilde{\mathbf{x}}$ the corrupted one. Following the standard abstraction of communication impairments, we model the corruption process in the order of additive Gaussian disturbance, bit-level corruption, and random missing.

First, continuous channel disturbance is modeled by additive Gaussian noise:

$$\mathbf{x}^{(1)} = \mathbf{x} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (22)$$

where σ controls the noise intensity. This step captures the physical-layer channel perturbation that is commonly modeled as additive white Gaussian noise (AWGN) in communication systems [6, 16, 36].

Second, discrete transmission errors are modeled as bit-level corruption:

$$x_i^{(2)} = \begin{cases} \phi^{-1}(\neg \phi(x_i^{(1)})), & i \in \mathcal{S}, \\ x_i^{(1)}, & i \notin \mathcal{S}, \end{cases} \quad (23)$$

where $\phi(\cdot)$ maps a scalar value to its encoded binary representation, $\neg$ denotes bitwise inversion, $\phi^{-1}(\cdot)$ maps the corrupted code back to the value space, and $\mathcal{S}$ is the set of corrupted positions. This step captures the discrete bit errors that arise after transmission and decoding. Unlike standard single-bit flipping, our implementation inverts all bits of a selected encoded value, resulting in sparse yet often large-magnitude deviations after decoding [13, 38, 41].

Finally, delivery-level failure is modeled by random missing:

$$\tilde{\mathbf{x}} = \mathbf{m} \odot \mathbf{x}^{(2)}, \quad m_i \sim \text{Bernoulli}(1 - p_{\text{miss}}), \quad (24)$$

where p_{miss} is the missing rate and $\odot$ denotes element-wise multiplication. This step captures incomplete feature delivery when lower-layer errors cannot be corrected reliably [7, 10, 23].

The above ordering is chosen to follow the communication pipeline. In practical transmission systems, analog channel disturbance is introduced first and is typically modeled by AWGN; after demodulation and decoding, such distortion is reflected in the bit error rate (BER); when these errors accumulate beyond the correction capability of the link, they further manifest as packet-level erasures or missing delivery.

5.3 Robustness Comparison

Under the standard GZSL setting, our method remains competitive with representative baselines, although QARM mainly targets robustness under corrupted feature conditions rather than clean-setting accuracy. As shown in Table ??, it still achieves performance comparable to strong baselines, particularly on UCF-GZSL, where the HM gap to the best reported result is less than 1%. Under corrupted conditions, however, existing methods suffer clearer performance drops, whereas our method remains more stable, indicating that the robustness-oriented design does not overly compromise standard discriminative capability.

Since evaluation under the proposed corruption setting requires re-running methods under a unified protocol, we compare against three representative baselines. As shown in Table 1, our method consistently performs best on all three benchmarks. The gains are most pronounced on UCF-GZSL, with 14.45/10.82/12.37 in S/U/HM, surpassing the strongest baseline by 7.08/2.31/4.50 points. Consistent with this observation, Fig. 5 shows that our method remains relatively stable as the missing rate increases, whereas the competing methods exhibit a much sharper performance drop after mild corruption. This trend is particularly clear in the Seen and HM metrics, further indicating that our method provides stronger robustness to missing-modality perturbations.

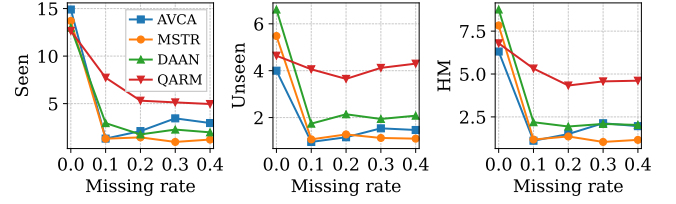


Figure 5: Performance across missing levels.

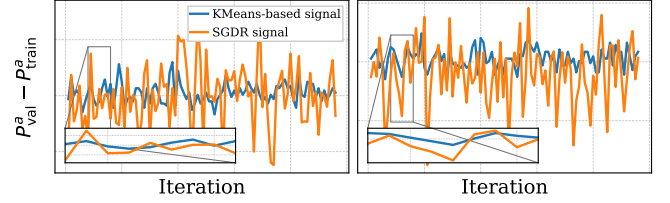


Figure 6: Branch-wise diagnostic discrepancy signals.

5.4 Ablation Studies of QARM

5.4.1 Robustness under Different Corruption Levels. We further evaluate robustness under different feature corruption levels, including both feature missingness and additive noise. As the missing rate increases, all methods show clear performance corruption, while QARM remains more stable and preserves stronger Seen and HM performance across most settings. A similar trend is also observed under increasing noise intensity, where the compared baselines deteriorate more rapidly but QARM maintains relatively stronger robustness. These results indicate that the proposed framework is effective against different types of feature-level corruption rather than only a single corruption pattern.

5.4.2 Analysis of OTDR. Table 2 reports the ablation results of OTDR. Removing either OT transport or the soft mask degrades performance, confirming that both components contribute to robust feature recovery. In particular, removing OT transport reduces HM from 12.37 to 12.04, suggesting that transport-guided latent alignment helps preserve structural consistency under corruption. The drop is larger when the soft mask is removed, where HM decreases to 11.43, indicating that soft mask fusion plays a more important role in stabilizing reconstruction from corrupted inputs. Overall, the full OTDR achieves the best result, supporting the effectiveness of combining both designs.

5.4.3 Analysis of SGDR. As shown in Fig. 6, the proposed SGDR produces a more responsive train-validation purity gap than the KMeans-based signal on both branches. In particular, local zoom-in views show that our signal varies more synchronously with branch-wise structural changes, indicating higher sensitivity to optimization inconsistency. By contrast, KMeans tends to yield smoother and less informative discrepancy patterns. This is mainly because, under high-dimensional sparse features, KMeans more easily forms coarse-grained partitions, and the resulting overlap statistics become overly averaged after masking, which weakens the ability of the signal to reflect subtle branch-wise differences. Therefore, the proposed discrepancy signal is more suitable for

Table 1: Comparison under different parameter settings.

Method	VGGSound-GZSL				UCF-GZSL				ActivityNet-GZSL			
	S	U	HM	ZSL	S	U	HM	ZSL	S	U	HM	ZSL
Missing rate = 0.1, AWGN intensity = 0.6, Bit-flip rate = 0.1												
AVCA[20]	1.32	0.96	1.11	2.09	4.17	7.76	5.42	12.68	2.42	0.88	1.29	3.43
MSTR[40]	1.31	1.07	1.18	2.23	7.89	6.19	6.93	12.60	4.24	1.25	1.93	4.06
DAAN[42]	2.95	1.74	2.19	2.55	9.38	8.58	8.96	14.86	3.47	2.51	2.92	3.38
QARM(ours)	7.73	4.06	5.32	5.01	15.02	13.54	14.24	17.26	4.75	3.40	3.96	3.64
Missing rate = 0.1, AWGN intensity = 0.7, Bit-flip rate = 0.1												
AVCA[20]	1.81	0.72	1.03	2.03	7.04	8.00	7.49	12.78	4.75	3.40	3.96	3.64
MSTR[40]	1.56	0.96	1.19	2.41	9.67	10.35	10.00	16.25	2.54	2.98	2.74	3.91
DAAN[42]	1.97	1.71	1.83	2.41	6.63	8.91	7.61	12.35	2.75	3.02	2.88	4.28
QARM(ours)	5.20	3.19	3.95	3.51	18.17	11.77	14.28	15.65	4.62	3.57	4.03	3.68
Missing rate = 0.1, AWGN intensity = 0.8, Bit-flip rate = 0.1												
AVCA[20]	3.45	1.54	2.13	3.01	5.41	5.50	5.45	12.14	2.26	2.01	2.13	2.98
MSTR[40]	0.97	1.13	1.04	2.58	12.76	6.97	9.02	15.73	2.82	2.44	2.62	3.63
DAAN[42]	1.76	2.14	1.94	2.43	6.68	8.29	7.40	11.87	2.78	2.55	2.66	3.60
QARM(ours)	3.08	3.28	3.18	3.54	13.11	13.30	13.20	16.80	4.95	3.37	4.01	3.62
Missing rate = 0.2, AWGN intensity = 0.7, Bit-flip rate = 0.1												
AVCA[20]	1.45	0.90	1.11	2.00	6.36	3.68	4.67	11.21	1.47	1.01	1.19	2.28
MSTR[40]	2.55	0.87	1.30	2.49	13.63	4.53	6.80	15.88	2.69	1.60	2.01	2.66
DAAN[42]	1.79	2.03	1.90	2.75	7.91	9.66	8.70	12.85	2.31	2.21	2.26	3.42
QARM(ours)	4.39	3.16	3.68	3.57	10.98	11.91	11.43	15.89	4.00	3.33	3.63	3.70
Missing rate = 0.2, AWGN intensity = 0.8, Bit-flip rate = 0.1												
AVCA[20]	2.61	1.45	1.86	2.67	4.91	5.26	5.08	11.48	2.08	0.23	0.41	2.77
MSTR[40]	2.86	1.21	1.70	2.98	6.56	4.60	5.41	12.66	1.13	0.90	1.00	2.61
DAAN[42]	2.00	1.91	1.81	2.46	7.15	8.34	7.69	11.87	2.61	2.35	2.47	3.44
QARM(ours)	5.72	2.93	3.87	3.45	15.04	9.89	10.60	10.24	3.25	3.30	3.28	3.43
Missing rate = 0.3, AWGN intensity = 0.8, Bit-flip rate = 0.1												
AVCA[20]	3.09	2.55	2.79	3.25	5.09	6.67	5.78	10.25	1.67	1.27	1.44	2.77
MSTR[40]	2.35	0.72	1.11	2.99	8.28	5.87	6.87	11.78	2.47	1.32	1.72	2.64
DAAN[42]	1.64	1.57	1.60	2.20	6.85	7.08	6.97	10.88	2.75	2.29	2.50	3.29
QARM(ours)	3.36	3.30	3.33	3.80	13.38	10.21	11.58	14.44	2.80	2.91	2.86	3.03
Average												
AVCA[20]	2.01	1.24	1.49	2.49	5.50	6.22	5.62	11.64	2.00	1.17	1.40	2.92
MSTR[40]	1.50	1.27	1.20	2.62	10.22	6.20	7.45	14.22	2.98	1.71	2.09	3.46
DAAN[42]	2.00	1.77	1.85	1.45	7.37	8.51	7.87	12.39	2.90	2.40	2.58	3.56
QARM(ours)	4.95	3.55	4.06	4.07	14.45	10.82	12.37	16.30	3.85	3.29	3.53	3.47

Table 2: OTDR ablation in corrupted GZSL.

Method	UCF-GZSL		
	S	U	HM
w/o OT Transport	12.53	11.80	12.04
w/o Soft Mask	10.98	11.91	11.43
Full OTDR	14.45	10.82	12.37

diagnostic control. To further isolate the effect of the diagnostic signal, we compare SGDR with a KMeans-based variant under the same backbone and corruption setting. As shown in Table 3, SGDR improves Seen, Unseen, HM, and ZSL from 4.00/3.33/3.63/3.70 to 4.71/3.66/4.12/3.90, respectively. The gain in HM is particularly notable, indicating that the proposed signal is more effective for balanced cross-modal transfer under corrupted conditions. These results confirm the effectiveness of the proposed SGDR signal.

As shown in Fig. 7, the audio and visual branches of AVCA exhibit a clear performance gap during training, with the visual branch

Table 3: Ablation study of the diagnostic signal.

Method	ActivityNet-GZSL			
	S	U	HM	ZSL
missing rate=0.2, AWGN intensity=0.2, Bit-flip=0.0				
KMeans [34]	4.00	3.33	3.63	3.70
SGDR	4.71	3.66	4.12	3.90

Table 4: Ablation of OTDR and SGDR on the MSTR backbone.

Method	S	U	HM
missing rate=0.2, AWGN intensity=0.6, Bit-flip=0.0			
MSTR	7.89	6.19	6.93
MSTR + OTDR	16.93	7.83	10.71
MSTR + SGDR	8.13	7.64	7.88
MSTR + OTDR + SGDR	17.45	15.38	16.35

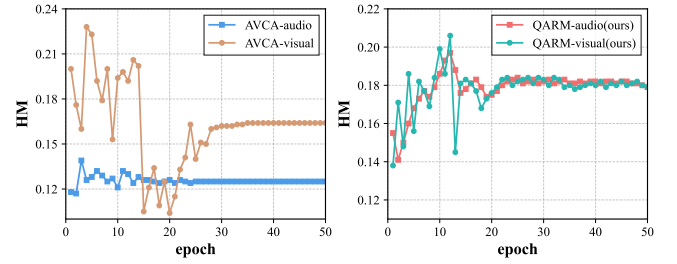
consistently outperforming the audio branch for a long period. This indicates modality imbalance, where one branch dominates while the other remains under-optimized. In contrast, after introducing the modality balancing mechanism, QARM produces much closer HM trajectories. Although slight asynchronous fluctuations still appear early on, both branches gradually converge to a similar and stable level. These results suggest that the proposed strategy alleviates modality imbalance, enhances branch-wise coordination, and promotes more balanced multimodal learning. As training proceeds, the stronger branch is explicitly suppressed to prevent persistent optimization dominance, thereby allowing the weaker branch to obtain more adequate learning. This gradually narrows the inter-branch gap and leads to more coordinated multimodal optimization. Such a trend further indicates that the improvement brought by QARM is rooted in a more balanced training dynamic rather than only a better final HM.

5.4.4 Transferability to Other Backbones. To further verify the transferability of the proposed modules, we integrate OTDR and SGDR into another representative backbone, namely MSTR, and report the results in Table 4. Under the same corrupted setting, the original MSTR achieves 6.93 HM. Adding OTDR improves HM to 10.71, while adding SGDR alone raises it to 7.88. When both modules are jointly introduced, the HM is further boosted to 16.35, with seen and unseen accuracies reaching 17.45 and 15.38, respectively. These results further indicate that the proposed modules are not restricted to AVCA, but can also be effectively and consistently transferred to other multimodal GZSL backbones.

5.4.5 Robustness under Asymmetric Cross-Modal Corruption. To further examine the robustness of QARM under modality-level information-quality inconsistency, we evaluate the proposed method on UCF-GZSL under two asymmetric cross-modal corruption settings. As reported in Table 5, QARM maintains relatively stable performance when the corruption severity is uneven across the audio and visual branches. Notably, under the *Audio severe*,

Table 5: Performance under asymmetric cross-modal corruption settings on UCF-GZSL.

S	U	HM	ZSL
<i>Audio: missing rate=0.1, Video: missing rate=0.1</i>			
14.45	10.82	12.37	16.30
<i>Audio: missing rate=0.1, Video: missing rate=0.4</i>			
12.13	9.94	10.92	15.31
<i>Audio: missing rate=0.4, Video: missing rate=0.1</i>			
17.02	9.71	12.37	16.55

**Figure 7: Comparison of harmonic mean (HM) trajectories of the audio and visual branches during training.**

Video mild setting, the HM remains identical to that of the symmetric setting. These results suggest that QARM is effective not only in handling feature corruption itself, but also in mitigating the adverse effect of cross-modal quality gaps on modality balance. This further indicates that the proposed framework can preserve balanced cross-modal transfer even when the two modalities are corrupted at different levels.

6 Conclusion

In this paper, we studied corrupted audio-visual generalized zero-shot learning in practical edge-cloud collaborative scenarios, where intermediate multimodal representations are frequently affected by noise, missingness, and transmission-induced distortion. To address this challenge, we proposed QARM, which integrates OTDR for robust feature reconstruction and SGDR for branch-wise diagnostic regulation within a unified framework. Experimental results on VGGSound-GZSL, UCF-GZSL, and ActivityNet-GZSL show that QARM not only remains competitive under the standard setting, but also achieves consistently stronger robustness under corrupted conditions. These results confirm the benefit of jointly improving representation fidelity and learning stability, and highlight the importance of corruption-aware multimodal modeling for robust audio-visual GZSL in realistic deployment settings. Future work will focus on more general corruption processes, stronger transmission-aware optimization strategies, and more generalizable multimodal transfer mechanisms for edge-cloud collaborative intelligence.

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